
HK IMO Training 2020-2021
Problem Set 5
For 21 November 2020



Problem 17. Find all triangular numbers which are also perfect squares.

(Note: A triangular number is a number of the form $1 + 2 + \cdots + n$.)

Problem 18. Prove that there exist infinitely many positive integers n such that $p = nr$, where p and r are respectively the semiperimeter and the inradius of a triangle with integer side lengths.

Problem 19. A *fissile square* is a positive integer which is a perfect square, and whose digits form two perfect squares in a row, which will be called the *left square* and the *right square*. For example, 49, 1444, and 1681 are fissile squares (decomposing as $4 \mid 9$, $144 \mid 4$, and $16 \mid 81$ respectively). Neither the left nor the right square may begin with the digit 0. Which perfect squares are the right square of infinitely many fissile squares?

Problem 20. Find all pairs of positive integers (m, n) such that $(2^m - 1)(2^n - 1)$ is a perfect square.